

COUNCIL BLUFFS, IA, USA

WMO: 725497

Lat: 41.259N

Lon: 95.760W

Elev: 1253

StdP: 14.04

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04908

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.8	2.6	-10.6	3.2	-0.2	-6.3	4.1	5.5	27.8	25.3	25.4	24.1	8.6	330	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	93.0	78.0	90.5	76.9	88.1	75.4	81.9	89.2	79.4	87.1	77.5	85.0	11.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.5	166.0	87.9	77.1	147.9	84.6	74.9	137.3	82.6	46.7	89.5	44.0	87.5	41.8	84.8	90.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
24.8	21.6	19.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-8.8	97.4	5.0	2.1	-12.4	98.9	-15.3	100.1	-18.1	101.3	-21.7	102.9
			-8.0	84.2	5.8	2.4	-12.2	85.9	-15.5	87.4	-18.8	88.7	-23.0	90.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	25.1	28.8	40.4	52.5	63.2	73.2	75.9	74.4	67.2	54.3	40.1	28.7	
	DBStd	20.62	12.13	12.34	12.38	10.04	8.75	6.27	5.86	5.60	7.94	9.33	10.71	11.87	
	HDD50	2867	773	597	346	88	7	0	0	0	2	62	326	666	
	HDD65	5851	1238	1014	764	390	141	9	2	4	68	346	747	1128	
	CDD50	3640	1	4	50	162	416	695	804	758	518	197	30	5	
	CDD65	1147	0	0	3	14	85	255	341	297	134	15	1	2	
	CDH74	10621	0	0	28	178	785	2381	3344	2548	1180	141	8	28	
	CDH80	3914	0	0	3	51	255	889	1321	958	397	28	0	12	
Wind	WSAvg	9.1	9.5	9.8	10.7	11.1	9.8	8.6	6.9	6.8	8.0	9.2	9.6	9.2	
Precipitation	PrecAvg	33.1	0.9	1.1	2.0	3.4	4.9	5.2	3.9	4.7	2.5	2.1	1.3	1.0	
	PrecMax	43.1	1.7	2.5	4.5	7.4	9.2	11.3	9.6	10.3	6.7	5.2	2.7	4.8	
	PrecMin	22.3	0.3	0.3	0.2	0.5	1.7	0.7	0.3	1.3	0.4	0.0	0.0	0.0	
	PrecStd	5.6	0.4	0.6	1.1	1.5	2.1	2.7	2.2	2.6	1.7	1.5	0.9	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.7	64.3	77.3	86.3	92.5	94.8	97.1	96.4	91.5	83.9	73.0	74.9	
		MCWB	45.2	52.6	59.0	65.0	71.9	75.2	80.6	78.3	74.6	66.4	58.2	64.7	
	2%	DB	50.3	57.1	71.7	79.0	86.2	91.0	92.8	91.3	89.5	78.7	66.1	54.3	
		MCWB	42.3	48.0	57.7	61.1	69.8	75.5	81.1	78.5	74.3	63.1	56.5	46.9	
	5%	DB	45.4	51.7	64.2	73.0	81.8	88.3	90.3	88.5	84.3	73.2	61.0	48.3	
		MCWB	39.4	44.7	54.1	58.7	67.4	74.3	79.0	76.7	72.1	60.6	51.1	42.9	
	10%	DB	40.9	45.5	59.2	68.4	78.6	85.6	87.6	85.7	81.2	70.0	55.3	43.7	
		MCWB	36.4	40.0	50.2	56.5	65.0	72.7	77.4	75.5	69.4	59.2	47.5	38.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.7	56.1	64.2	67.5	75.5	81.7	85.9	83.9	78.7	71.0	62.3	65.0
			MCDB	55.1	59.6	72.6	76.4	84.4	87.3	93.4	92.0	88.0	79.0	68.9	74.3
2%		WB	42.8	49.0	60.0	64.2	72.7	78.4	82.9	80.5	76.1	66.8	57.3	48.2	
		MCDB	49.9	56.6	67.5	75.6	82.3	86.7	90.2	88.2	84.6	74.0	63.7	51.9	
5%		WB	39.6	44.2	55.6	61.2	70.2	76.5	80.3	78.4	74.1	63.7	52.9	42.7	
		MCDB	44.5	50.9	63.4	70.9	78.6	85.3	87.8	86.1	81.9	69.6	60.5	48.0	
10%		WB	36.7	40.2	50.5	58.2	67.6	74.6	78.1	76.4	71.9	60.2	48.6	39.1	
		MCDB	41.1	45.3	59.4	66.7	76.0	83.4	85.0	83.8	79.1	67.8	55.7	43.6	
Mean Daily Temperature Range	5% DB	MDBR	16.1	16.5	19.4	20.4	19.5	19.0	18.1	18.5	20.9	20.0	17.8	15.0	
		MCDBR	22.5	24.4	27.7	28.2	23.9	20.8	19.6	21.5	24.1	25.4	25.0	22.5	
		MCWBR	16.4	16.3	16.0	15.3	12.1	11.0	11.9	12.2	12.5	14.0	15.4	17.0	
	5% WB	MCDBR	21.2	22.9	24.5	24.6	21.4	19.1	18.2	19.4	21.1	21.3	23.5	21.0	
		MCWBR	15.8	16.0	15.3	15.1	12.4	12.2	12.4	12.8	12.6	14.5	16.4	16.7	
Clear-Sky Solar Irradiance	taub	0.279	0.313	0.327	0.378	0.402	0.406	0.420	0.408	0.370	0.331	0.296	0.282		
	taud	2.455	2.375	2.402	2.279	2.281	2.321	2.275	2.319	2.383	2.481	2.505	2.457		
	Ebn at Noon	281	284	293	283	278	276	270	271	274	275	272	269		
	Edh at Noon	24	30	33	41	42	40	42	39	34	27	23	22		
All-Sky Solar Radiation	RadAvg	634	934	1210	1542	1836	2038	2056	1776	1469	1023	710	534		
	RadStd	56	56	110	109	140	161	137	115	96	121	50	53		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page